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BIOFOULING HULL INSPECTION

HIFOFUA

FRANMARINE

WEDNESDAY 19TH AUGUST 2026

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EXECUTIVE SUMMARY

Hifofua, a 27 metre tug operated by Ports Authority Tonga, was inspected alongside Faua Wharf at Nuku'alofa on 19 August 2026. The inspection was conducted using a Deep Trekker Pivot remotely operated vehicle deployed from the wharf, and findings were recorded in the MarineStream platform. Fouling was assessed against the Level of Fouling (LoF) scale set out in the New Zealand Craft Risk Management Standard.

Underwater visibility was poor throughout. Fouling density prevented assessment of the antifouling coating on every area inspected and prevented resolution of the organisms present.

Inspection Findings

- **Rudder assembly:** LoF 5 across the top of the rudder, leading edge, both flat plates, both rudder hinges and both pintle bearings, with macrofouling covering 41 to 100 per cent of visible surfaces. Both inspection ports and the anodes rated LoF 4 at 16 to 40 per cent coverage. Anodes heavily fouled but approximately 80 per cent intact.
- **Propeller:** boss and all four blades rated LoF 5, with macrofouling covering the full visible surface. Calcareous fouling and heavy filamentous algae present. The propeller tunnel was heavily fouled. Fixings on the boss and blades were obscured and could not be sighted.
- **Stern arrangement:** rope guard, eddy cover, stern tube and A-bracket all rated LoF 5. The rope guard inspection ports could not be located.
- **Fouling character:** a mature composite community, consistent with a vessel held largely stationary in warm, sheltered harbour water, comprising a filamentous algal turf over a base of calcareous hard fouling with erect soft fouling and encrusting colonial growth across the sheltered, low-flow surfaces.

Areas Not Assessed

No fouling ratings were recorded for the hull plating forward, midships or aft, the bow and transom plates, skeg, bilge keels, draft marks, transducers, overboard discharges, or any of the five sea chests, external or internal. No antifouling coating condition was recorded anywhere on the vessel.

Conclusion

No invasive marine species determination can be made from this inspection in either direction. Fouling density and water clarity prevented identification of the organisms present on the areas assessed, and no data was recorded for the hull plating, sea chests and remaining niche areas.

1. PROJECT PARTICULARS

PROJECT:	Biofouling Hull Inspection	
VESSEL DETAILS:	**Vessel:**	Hifofua
	IMO:	9109213
	Commissioned:	1994
	Gross Tonnage (t):	180
	Length (m):	27 m
	Beam (m):	8 m
	Draft (m):	
DATE OF INSPECTION:	Wednesday 19 August 2026	
CLIENT:	Ports Authority Tonga	
JOB LOCATION:	Nuku'alofa , Faua Wharf	
INSPECTION TEAM:	Supervisor:	Shannon Ward
	ROV Unit:	Deep Trekker Pivot
	Inspector:	Marc Green

2. OVERVIEW

Franmarine was engaged to conduct a biofouling hull inspection of the tug *Hifofua*, operated by Ports Authority Tonga, alongside Faua Wharf at Nuku'alofa. *Hifofua* is a 27 metre, 180 gross tonne tug built in 1994, Tongan flagged and registered at Nuku'alofa under IMO 9109213.

The inspection was undertaken to assess the condition of the vessel's underwater surfaces, record the level of fouling present across the hull and niche areas, and identify any potential invasive marine species.

The scope of work and work method are based on the following guidance:

- The Craft Risk Management Standard for Biofouling on Vessels Arriving to New Zealand, New Zealand Ministry for Primary Industries
- 2023 Guidelines for the control and management of ships' biofouling to minimise the transfer of invasive aquatic species (Resolution MEPC.378(80))

Franmarine has developed MarineStream, a biofouling management platform, to provide a secure and immutable record book that maintains compliance with the requirements listed above.

3. METHODOLOGY

Franmarine was directed to conduct a biofouling hull inspection of *Hifofua* at Faua Wharf, Nuku'alofa, covering hull plating, the bow and transom plates, sea chests, overboard discharges, transducers, draft marks, bilge keels, skeg, stern arrangement, rudder and propeller.

The inspection was conducted using a remotely operated vehicle deployed from the wharf.

Inspection findings were recorded in the MarineStream technology platform, ensuring inspection processes are standardised and an immutable record of the findings is maintained. Biofouling was assessed using the Craft Risk Management Standard for Biofouling on Vessels Arriving to New Zealand, issued by the Ministry for Primary Industries, specifically the Level of Fouling (LoF) definitions and typical examples. The condition of the antifouling coating was also to be assessed.

Documents used to assist the project include:

- Franmarine Operations Manual
- Client supplied vessel drawings
- The Craft Risk Management Standard, New Zealand Ministry for Primary Industries

4. REFERENCE TABLES

The below tables are referenced throughout the report to formally standardise both biofouling and antifouling condition ratings referred to herein.

4.1 LEVEL OF FOULING (LOF) SCALE

LEVEL OF FOULING	DESCRIPTION
Rank: 0	0% No slime layer. No macrofouling. Only clean surfaces.
Rank: 1	0% Slime Layer on some or all surfaces. No macrofouling.
Rank: 2	1-5% of visible surfaces - Macrofouling present in small patches or a few isolated individuals or small colonies.
Rank: 3	6-15% of visible surfaces - Considerable macrofouling on surfaces.
Rank: 4	16-40% of visible surfaces - Extensive macrofouling present but more than hald of surfaces without biofouling.
Rank: 5	41-100% of visible surfaces - Very heavy macrofouling present covering substantial portions of visible surfaces.

4.2 PAINT DETERIORATION RATING (PDR) SCALE

RATING	DESCRIPTION
PDR: 0	No anti-foul (AF) coating applied.
PDR: 10	Anti-Foul (AF) Paint intact.
PDR: 20	AF paint missing from edges, corners, seams, welds, rivet, or bolt heads to expose anti corrosion (AC) paint. (undercoat).
PDR: 30	AF paint missing from slightly curved or flat areas to expose underlying AC paint; An AF paint with visible brush swirl marks within the outermost layer – not extending into underlying layers of paint.
PDR: 40	AF paint missing from intact blisters to expose AC paint or an AF coating with visible brush swirl marks exposing the next underlying layer of AF or AC paint.
PDR: 50	AF blisters ruptured to expose intact AC paint.
PDR: 60	AF/AC paint missing or peeling to expose steel substrate, no corrosion present.
PDR: 70	AF/AC paint removed from edges, corners, seams, welds, rivet or bolt heads to expose steel substrate with corrosion present.
PDR: 80	Ruptured AF/AC blisters on slightly curved or flat surfaces with corrosion or corrosion stains present.
PDR: 90	Area corrosion of steel substrate with no AF/AC paint cover due to peeling or abrasion damage.
PDR: 100	Area corrosion showing visible surface evidence of pitting, scaling, and roughening of steel substrate.

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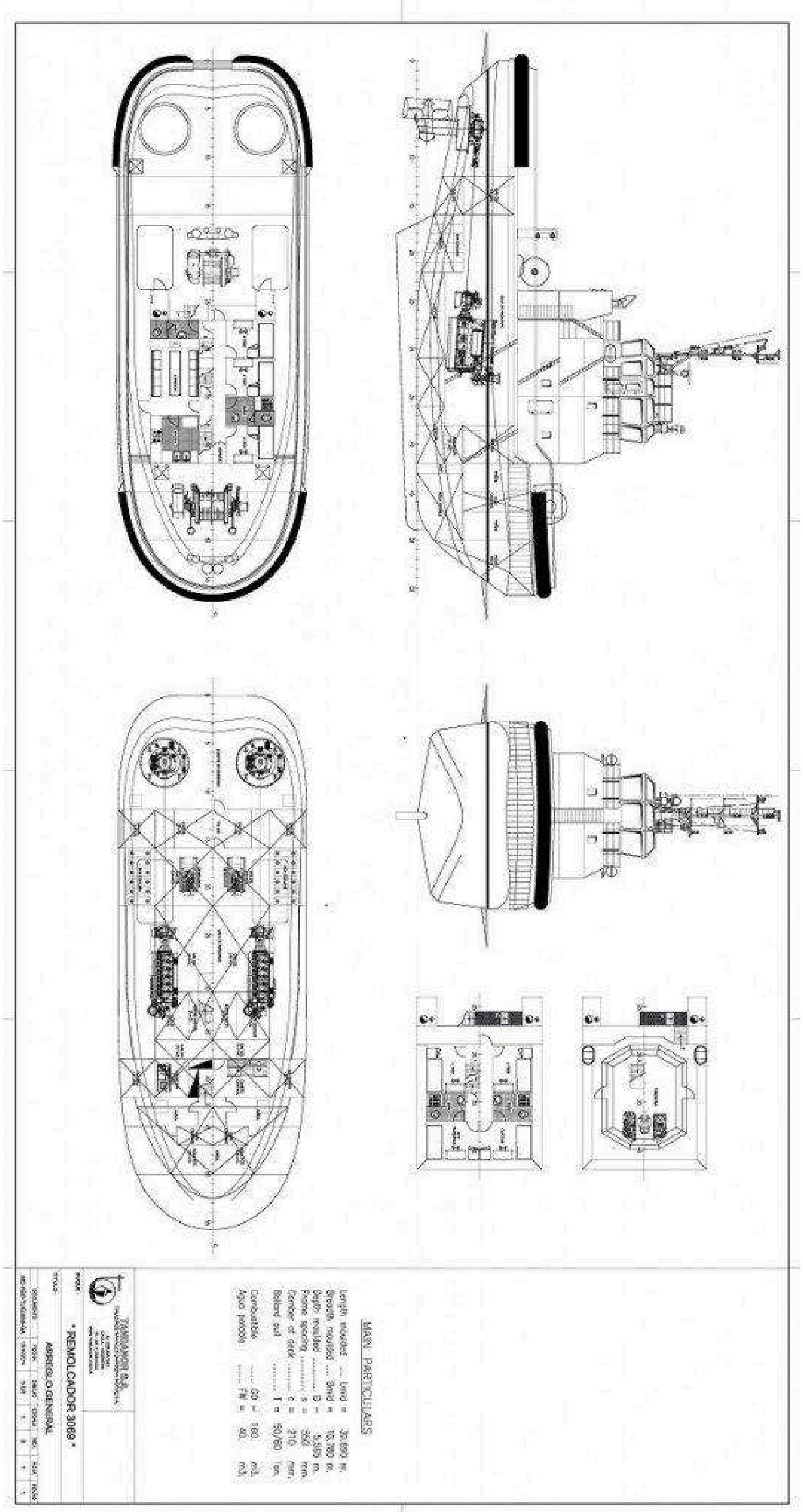
5. GENERAL ARRANGEMENT



5.1 RUDDER

5.1.1 OBSERVATIONS AND FINDINGS

Heavy fouling recorded across the majority of the rudder. Top of rudder, leading edge, both rudder flat plates, both rudder hinges and both pintle bearings all rated LoF 5, with macrofouling covering 41–100 per cent of visible surfaces. Both inspection ports and the anodes rated LoF 4 at 16–40 per cent coverage. Anodes heavily fouled but appear approximately 80 per cent intact. Fouling density prevented assessment of the underlying coating.

Description	Level of fouling (LOF)	PDR Rating	Comments
Top of Rudder	Rank 5 | - 41-100% of visible surfaces - Very heavy macrofouling present covering substantial portions of visible surfaces.	No PDR rating allocated	No additional comments
Pintle Bearing - Port	Rank 5 | - 41-100% of visible surfaces - Very heavy macrofouling present covering substantial portions of visible surfaces.	No PDR rating allocated	No additional comments
Pintle Bearing - Starboard	Rank 5 | - 41-100% of visible surfaces - Very heavy macrofouling present covering substantial portions of visible surfaces.	No PDR rating allocated	No additional comments
Rudder Hinge - Port	Rank 5 | - 41-100% of visible surfaces - Very heavy macrofouling present covering substantial portions of visible surfaces.	No PDR rating allocated	No additional comments
Rudder Hinge - Starboard	Rank 5 | - 41-100% of visible surfaces - Very heavy macrofouling present covering substantial portions of visible surfaces.	No PDR rating allocated	No additional comments
Leading Edge	Rank 5 | - 41-100% of visible surfaces - Very heavy macrofouling present covering substantial portions of visible surfaces.	No PDR rating allocated	No additional comments
Rudder Flat Plate - Port	Rank 5 | - 41-100% of visible surfaces - Very heavy macrofouling present covering substantial portions of visible surfaces.	No PDR rating allocated	No additional comments

Description	Level of fouling (LOF)	PDR Rating	Comments
Rudder Flat Plate - Starboard	Rank 5 | - 41-100% of visible surfaces - Very heavy macrofouling present covering substantial portions of visible surfaces.	No PDR rating allocated	No additional comments
Inspection Port - Port	Rank 4 | - 16-40% of visible surfaces - Extensive macrofouling present but more than half of surfaces without biofouling.	No PDR rating allocated	No additional comments
Inspection Port - Starboard	Rank 4 | - 16-40% of visible surfaces - Extensive macrofouling present but more than half of surfaces without biofouling.	No PDR rating allocated	No additional comments
Anodes	Rank 4 | - 16-40% of visible surfaces - Extensive macrofouling present but more than half of surfaces without biofouling.	No PDR rating allocated	No additional comments

INSPECTOR COMMENTS

The fouling is a mature composite community, consistent with a vessel held largely stationary in warm, sheltered harbour water. A filamentous algal turf overlies a base of calcareous hard fouling, with erect soft fouling and encrusting colonial growth established across the sheltered, low-flow surfaces. Fouling density and poor visibility prevented resolution of individual organisms, so composition is described at category level only. Confirmation would require physical sampling and taxonomic identification.

5.2 PROPELLER

5.2.1 OBSERVATIONS AND FINDINGS

Propeller inspected by ROV in poor visibility. Boss and all four blades rated LoF 5, with macrofouling covering the full visible surface. Calcareous fouling and heavy filamentous algae present. Fixings on the boss and blades were not visible due to the level of fouling. Propeller tunnel heavily fouled with calcareous fouling and filamentous algae.

Description	Level of fouling (LOF)	PDR Rating	Comments
Boss	Rank 5 | - 41-100% of visible surfaces - Very heavy macrofouling present covering substantial portions of visible surfaces.	No PDR rating allocated	No additional comments
Blade 1	Rank 5 | - 41-100% of visible surfaces - Very heavy macrofouling present covering substantial portions of visible surfaces.	No PDR rating allocated	No additional comments
Blade 2	Rank 5 | - 41-100% of visible surfaces - Very heavy macrofouling present covering substantial portions of visible surfaces.	No PDR rating allocated	No additional comments
Blade 3	Rank 5 | - 41-100% of visible surfaces - Very heavy macrofouling present covering substantial portions of visible surfaces.	No PDR rating allocated	No additional comments
Blade 4	Rank 5 | - 41-100% of visible surfaces - Very heavy macrofouling present covering substantial portions of visible surfaces.	No PDR rating allocated	No additional comments

INSPECTOR COMMENTS

The fouling is a mature composite community, consistent with a vessel held largely stationary in warm, sheltered harbour water. A filamentous algal turf overlies a base of calcareous hard fouling, with erect soft fouling and encrusting colonial growth established across the sheltered, low-flow surfaces. Fouling density and poor visibility prevented resolution of individual organisms, so composition is described at category level only. Confirmation would require physical sampling and taxonomic identification.








5.3 STERN ARRANGEMENT

5.3.1 OBSERVATIONS AND FINDINGS

Stern arrangement inspected by ROV in poor visibility. Rope guard, eddy cover, stern tube and A-bracket all rated LoF 5, with macrofouling covering 41–100 per cent of visible surfaces. Coating condition could not be assessed due to the level of fouling. The rope guard inspection ports could not be located.

Description	Level of fouling (LOF)	PDR Rating	Comments
Rope Guard	Rank 5 | - 41-100% of visible surfaces - Very heavy macrofouling present covering substantial portions of visible surfaces.	No PDR rating allocated	No additional comments
Eddy Cover	Rank 5 | - 41-100% of visible surfaces - Very heavy macrofouling present covering substantial portions of visible surfaces.	No PDR rating allocated	No additional comments
Stern Tube	Rank 5 | - 41-100% of visible surfaces - Very heavy macrofouling present covering substantial portions of visible surfaces.	No PDR rating allocated	No additional comments
A Bracket	Rank 5 | - 41-100% of visible surfaces - Very heavy macrofouling present covering substantial portions of visible surfaces.	No PDR rating allocated	No additional comments

INSPECTOR COMMENTS

The fouling is a mature composite community, consistent with a vessel held largely stationary in warm, sheltered harbour water. A filamentous algal turf overlies a base of calcareous hard fouling, with erect soft fouling and encrusting colonial growth established across the sheltered, low-flow surfaces. Fouling density and poor visibility prevented resolution of individual organisms, so composition is described at category level only. Confirmation would require physical sampling and taxonomic identification.

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6. RECOMMENDATIONS

- Complete the inspection record. Forward, midships and aft hull plating, the bow and transom plates, skeg, bilge keels, draft marks, transducers, overboard discharges and all five sea chests remain unrated. The report cannot support a compliance position until these are captured, or until the reason they could not be inspected is recorded.
- Undertake an in-water hull clean with capture, prioritising the rudder assembly, propeller and stern arrangement.
- Collect samples during cleaning so the organisms present can be identified and the invasive marine species question closed out. This inspection cannot answer it.
- Reassess antifouling coating condition once fouling is removed. No coating data exists for any part of the vessel.
- Verify the integrity of the propeller boss and blade fixings once cleaned, and locate and inspect the rope guard inspection ports.
- Confirm cathodic protection performance once the anodes are cleared. Anodes are approximately 80 per cent intact and were heavily fouled at inspection.
- Create the vessel's Biofouling Management Plan.
- Establish a six-month inspection interval following cleaning to monitor reaccumulation.

Please be advised that CCTV imagery and additional still photographs may be provided upon client request.

Thank you for the opportunity to provide our services.

If we can be of any future assistance, please do not hesitate to contact us at any time.

Supervisor Declaration
I, Shannon Ward, declare my inspection and findings outlined above to be true and correct to the best of my knowledge on the date of inspection. **Signed:** **Date:** Wednesday 19 August 2026

IMS Declaration
I, Marc Green, declare my inspection and findings outlined above to be true and correct to the best of my knowledge on the date of inspection. PP MH **Signed:** **Date:** Wednesday 19 August 2026

END OF REPORT

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Document Status		
Revision: 1	**Document Owner:** Adam Falconer-West	**Document Number:** 1
Action	**Name and Position**	**Date**
Prepared By	MarineStream	Wednesday 19 August 2026
Reviewed By	Mathew Harvey	Wednesday 19 August 2026
Approved By	Mathew Harvey	Wednesday 19 August 2026